北京交通大學

信号与系统

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and Information Engineering

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System Frequency Domain Analysis

- Why Perform Frequency Domain Analysis of a System
- Frequency Domain Analysis of Continuous Time LTI Systems
- Frequency Domain Analysis of Discrete Time LTI Systems
- Signal Amplitude Modulation and Demodulation
- Frequency Domain Analysis of System Using MATLAB

Frequency Domain Analysis of Discrete Time LTI Systems

- ※ Frequency Domain Description of Discrete Time LTI System
- ※ Frequency Domain Analysis of Discrete Time LTI System Response
- ※ ideal digital filter

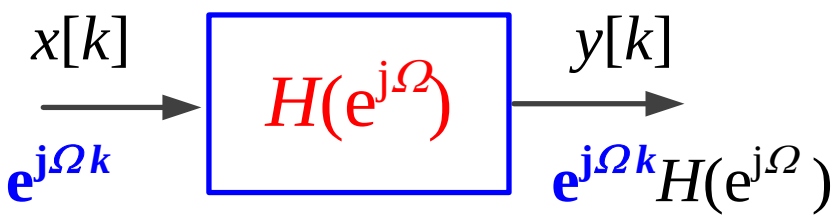
Frequency Domain Analysis of Discrete Time LTI Systems

- Response of an imaginary exponential signal $e^{j\Omega k}$ through a discrete LTI system

$$y[k] = e^{j\Omega k} * h[k] = \sum_n e^{j\Omega(k-n)} h[n]$$

$$= e^{j\Omega k} \underbrace{\sum_n e^{-j\Omega n} h[n]}_{H(e^{j\Omega})}$$

$$= e^{j\Omega k} H(e^{j\Omega})$$



Frequency Domain Analysis of Discrete Time LTI Systems

- **Frequency response $H(e^{j\Omega})$**

$$H(e^{j\Omega}) = \mathrm{DTFT}\{h[k]\} = \sum_{k=-\infty}^{\infty} h[k]e^{-j\Omega k}$$

- **Magnitude Response vs. Phase Response**

$$H(e^{j\Omega}) = |H(e^{j\Omega})|\, e^{j\varphi(\Omega)}$$

magnitude response

phase response

If $h[k]$ is a real signal, the amplitude response $|H(e^{j\Omega})|$ is an even function of Ω,

The phase response $\varphi(\Omega)$ is an odd function of Ω.

Frequency Domain Description of Discrete Time LTI System

➢ Another Definition of Frequency Response $H(e^{j\Omega})$

Frequency Domain Representation of Zero State Responses

$$y_{zs}[k] = x[k] * h[k]$$

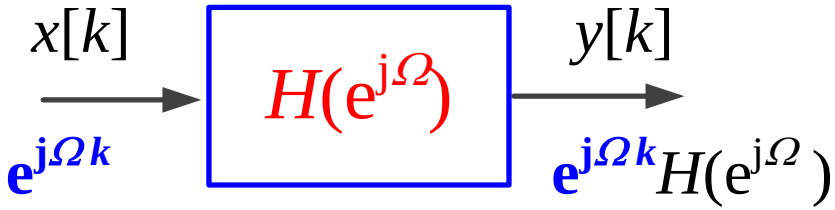
Using DTFT's **Convolution characteristics**

$$Y_{zs}(e^{j\Omega}) = X(e^{j\Omega})H(e^{j\Omega})$$

$$H(e^{j\Omega}) = \frac{Y_{zs}(e^{j\Omega})}{X(e^{j\Omega})}$$

Frequency Domain Description of Discrete Time LTI System

➢ The physical meaning of $H(e^{j\Omega})$



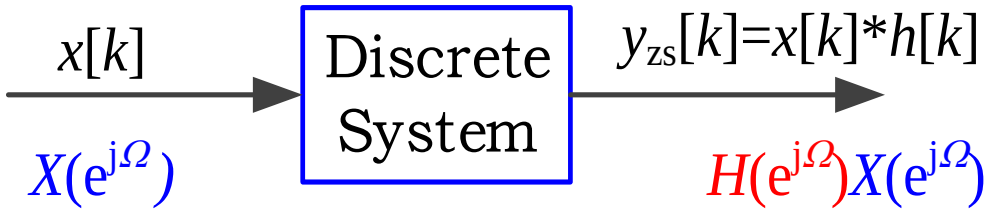
Response of an imaginary exponential signal $e^{j\Omega k}$ at frequency Ω through a discrete LTI system

It is still an imaginary exponential signal with the same frequency. The change of signal is determined by $H(e^{j\Omega})$.

$H(e^{j\Omega})$Reflects the ***transmission characteristics*** of the discrete system for imaginary exponential signals of different frequencies。

Frequency Domain Description of Discrete Time LTI System

- **the physical meaning of $H(e^{j\Omega})$**



The discrete LTI system changes the input signal spectrum $X(e^{j\Omega})$ into $H(e^{j\Omega})\,X(e^{j\Omega})$, and the change rule is completely determined by $H(e^{j\Omega})$.

$H(e^{j\Omega})$ Reflects the transmission characteristics of the discrete system for imaginary exponential signals of different frequencies

Frequency Domain Description of Discrete Time LTI System

- **General Form of Frequency Response $H(e^{j\Omega})$ of Discrete Time LTI System**

If the difference equation describing the discrete LTI system is

$$y[k]+a_1y[k-1]+\cdots+a_{n-1}y[k-n+1]+a_ny[k-n]=$$

$$b_0x[k]+b_1x[k-1]+\cdots+b_{m-1}x[k-m-1]+b_mx[k-m]$$

Using the DTFT displacement characteristics, the frequency domain equation describing the system can be obtained

$$[1+a_1\,\mathrm{e}^{-\mathrm{j}\Omega}+\cdots+a_n\,\mathrm{e}^{-\mathrm{j}n\Omega}]Y_{\mathrm{zs}}(\mathrm{e}^{\mathrm{j}\Omega})=$$

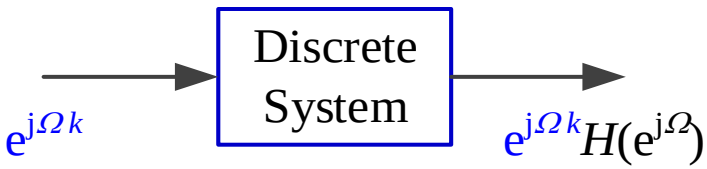
$$[b_0+b_1\,\mathrm{e}^{-\mathrm{j}\Omega}+\cdots+a_m\,\mathrm{e}^{-\mathrm{j}m\Omega}]X(\mathrm{e}^{\mathrm{j}\Omega})$$

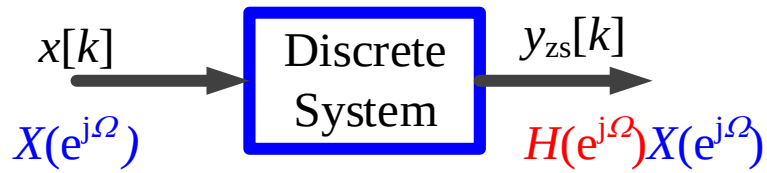
From the definition of the frequency response of a discrete system, we can get

$$H(\mathrm{e}^{\mathrm{j}\Omega})=\frac{Y_{\mathrm{zs}}(\mathrm{e}^{\mathrm{j}\Omega})}{X(\mathrm{e}^{\mathrm{j}\Omega})}=\frac{b_0+b_1\mathrm{e}^{-\mathrm{j}\Omega}+\cdots+b_{m-1}\mathrm{e}^{-\mathrm{j}\Omega(m-1)}+b_m\mathrm{e}^{-\mathrm{j}\Omega m}}{1+a_1\mathrm{e}^{-\mathrm{j}\Omega}+\cdots+a_{n-1}\mathrm{e}^{-\mathrm{j}\Omega(n-1)}+a_n\mathrm{e}^{-\mathrm{j}\Omega n}}$$

Frequency Domain Analysis of Discrete Time LTI Systems

- Response of non-periodic sequence through discrete LTI system





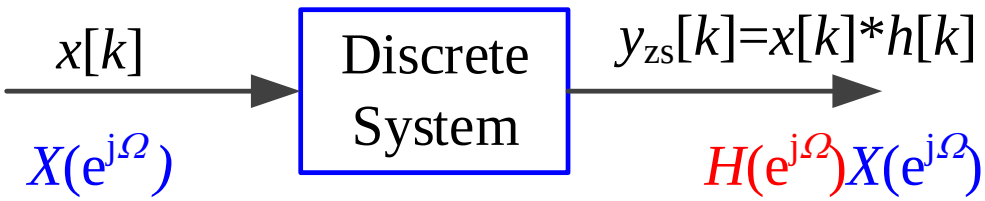
$$x[k] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\Omega})\underline{e^{j\Omega k}}\,d\Omega$$

$$y_{zs}[k] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\Omega})\underline{H(e^{j\Omega})e^{j\Omega k}}\,d\Omega$$

$$Y_{zs}(e^{j\Omega}) = H(e^{j\Omega})X(e^{j\Omega})$$

Frequency Domain Analysis of Discrete Time LTI Systems

➢The relationship between the **system zero state response frequency domain analysis method** and the **convolution integration method**:



The two analysis methods are essentially the same, but **the basic signal for expressing the signal is different**.

The **time-domain convolution theorem** of DTFT is a bridge connecting the two.

Frequency Domain Analysis of Discrete Time LTI Systems

Example: It is known that the difference equation describing a discrete-time LTI system is

$$y[k] - 0.75y[k-1] + 0.125y[k-2] = 4x[k]+3x[k-1],$$

Input $x[k] = (0.75)^k u[k]$, find the zero-state response $y_{zs}[k]$ of the system.

Solution: Perform DTFT on both sides of the difference equation to get

$$(1 - 0.75\mathrm{e}^{-\mathrm{j}\Omega} + 0.125\mathrm{e}^{-\mathrm{j}2\Omega})Y_{zs}(\mathrm{e}^{\mathrm{j}\Omega}) = (4 + 3\mathrm{e}^{-\mathrm{j}\Omega})X(\mathrm{e}^{\mathrm{j}\Omega})$$

According to the definition of the frequency response of a discrete system

$$H(\mathrm{e}^{\mathrm{j}\Omega}) = \frac{Y_{zs}(\mathrm{e}^{\mathrm{j}\Omega})}{X(\mathrm{e}^{\mathrm{j}\Omega})} = \frac{4 + 3\mathrm{e}^{-\mathrm{j}\Omega}}{1 - 0.75\mathrm{e}^{-\mathrm{j}\Omega} + 0.125\mathrm{e}^{-\mathrm{j}2\Omega}}$$

Frequency Domain Analysis of Discrete Time LTI Systems

Example: It is known that the difference equation describing a discrete-time LTI system is

$$y[k]-0.75y[k-1]+0.125y[k-2]=4x[k]+3x[k-1],$$

Input $x[k]=(0.75)^k u[k]$, find the zero-state response $y_{zs}[k]$ of the system.

Solution: The spectrum function $y_{zs}[k]$ of the zero-state response $Y_{zs}(e^{j\Omega})$ of the system is

$$Y_{zs}(e^{j\Omega})=H(e^{j\Omega})X(e^{j\Omega})=\frac{4+3e^{-j\Omega}}{(1-0.25e^{-j\Omega})(1-0.5e^{-j\Omega})(1-0.75e^{-j\Omega})}$$

$$=\frac{8}{1-0.25e^{-j\Omega}}+\frac{-40}{1-0.5e^{-j\Omega}}+\frac{4.5}{1-0.75e^{-j\Omega}}$$

Perform IDTFT on $Y_{zs}(e^{j\Omega})$ to get

$$y_{zs}[k]=8(0.25)^k u[k]-40(0.5)^k u[k]+4.5(0.75)^k u[k]$$

Frequency Domain Analysis of Discrete Time LTI Systems

- **Cosine signal $\cos(\Omega_0 k+\theta)$ through the response of the system**

From Euler's formula, it can be obtained

$$\cos(\Omega_0 k+\theta) = (\mathrm{e}^{\mathrm{j}(\Omega_0 k+\theta)} + \mathrm{e}^{-\mathrm{j}(\Omega_0 k+\theta)})/2$$

Using the characteristics of the imaginary exponential signal $\mathrm{e}^{\mathrm{j}\Omega_0 k}$ response and the linearity of the system, we can get

$$y[k] = \left[H(\mathrm{e}^{\mathrm{j}\Omega_0})\mathrm{e}^{\mathrm{j}(\Omega_0 k+\theta)} + H(\mathrm{e}^{-\mathrm{j}\Omega_0})\mathrm{e}^{-\mathrm{j}(\Omega_0 k+\theta)} \right]/2$$

$$\left|H(\mathrm{e}^{\mathrm{j}\Omega_0})\right|\mathrm{e}^{\mathrm{j}\varphi(\Omega_0)} \qquad \left|H(\mathrm{e}^{\mathrm{j}\Omega_0})\right|\mathrm{e}^{-\mathrm{j}\varphi(\Omega_0)}$$

$$= \left|H(\mathrm{e}^{\mathrm{j}\Omega_0})\right|\cos(\Omega_0 k + \varphi(\Omega_0) + \theta)$$

Frequency Domain Analysis of Discrete Time LTI Systems

- **Cosine signal $\cos(\Omega_0 k+\theta)$ through the response of the system**

$$\cos(\Omega_0 k+\theta) \longrightarrow \left|H(e^{j\Omega_0})\right| \cos\left[\Omega_0 k+\varphi(\Omega_0)+\theta\right]$$

The response of a cosine signal with frequency Ω_0 through a discrete LTI system

It is still a cosine signal with the same frequency.

The amplitude change of the response is determined by $|H(e^{j\Omega_0})|$,

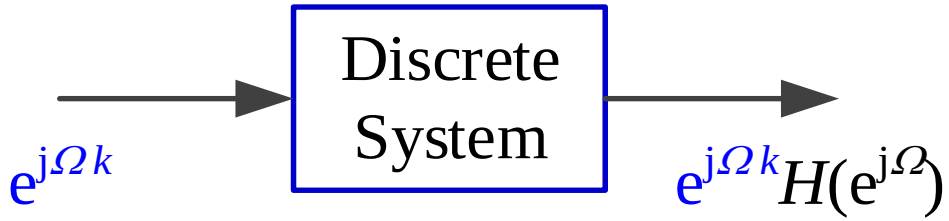
The phase change of the response is determined by $\varphi(\Omega_0)$.

In the same way, it can be launched

$$\sin(\Omega_0 k+\theta) \longrightarrow \left|H(e^{j\Omega_0})\right| \sin\left[\Omega_0 k+\varphi(\Omega_0)+\theta\right]$$

Frequency Domain Analysis of Discrete Time LTI Systems

- Response of a periodic sequence through an LTI system



$$\tilde{x}[k] = \frac{1}{N}\sum_{m=<N>} \tilde{X}[m]\, \underline{e^{j\frac{2\pi m}{N}k}}$$

$$\tilde{y}[k] = \frac{1}{N}\sum_{m=<N>} \tilde{X}[m]\, \underline{H(e^{j\frac{2\pi m}{N}})e^{j\frac{2\pi m}{N}k}}$$

Frequency Domain Analysis of Discrete Time LTI Systems

Example: $h[k]=(0.5)^k u[k]$ of a discrete LTI system is known, input $x[k]=\cos(0.5\pi k)$, $(-\infty<k<\infty)$, and find the steady-state response of the system.

Solution:

$$H(\mathrm{e}^{\mathrm{j}\Omega}) = \mathrm{DTFT}\{h[k]\} = \frac{1}{1-0.5\mathrm{e}^{-\mathrm{j}\Omega}}$$

$$H(\mathrm{e}^{\mathrm{j}0.5\pi}) = \frac{1}{1+0.5\mathrm{j}} = 0.8944\mathrm{e}^{-\mathrm{j}0.4636}$$

$$y[k] = \left|H(\mathrm{e}^{\mathrm{j}0.5\pi})\right| \cos[0.5\pi k + \varphi(0.5\pi)]$$

$$= 0.8944\cos(0.5\pi k - 0.4636)$$

Frequency Domain Analysis of Discrete Time LTI Systems

- Discrete-time LTI system with linear phase

$$H(\mathrm{e}^{\mathrm{j}\Omega}) = \left|H(\mathrm{e}^{\mathrm{j}\Omega})\right| \mathrm{e}^{\mathrm{j}\varphi(\Omega)} \quad \varphi(\Omega) = -\Omega k_0$$

linear phase system

$$x[k] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\mathrm{e}^{\mathrm{j}\Omega})\,\mathrm{e}^{\mathrm{j}\Omega k}\,\mathrm{d}\Omega$$

$$\boxed{\left|H(\mathrm{e}^{\mathrm{j}\Omega})\right| \mathrm{e}^{-\mathrm{j}\Omega k_0}}$$

$$y[k] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\mathrm{e}^{\mathrm{j}\Omega})\,|H(\mathrm{e}^{\mathrm{j}\Omega})|\,\mathrm{e}^{\mathrm{j}\Omega(k-k_0)}\,\mathrm{d}\Omega$$

When a signal passes through a linear-phase LTI system, only the amplitude is changed and the phase is not distorted.

Frequency Domain Analysis of Discrete Time LTI Systems

- **Discrete-time LTI system with linear phase**

$$H(\mathrm{e}^{\mathrm{j}\Omega}) = \left|H(\mathrm{e}^{\mathrm{j}\Omega})\right|\mathrm{e}^{\mathrm{j}\varphi(\Omega)} = -\Omega k_0$$

linear phase system

$$\tilde{x}[k] = \frac{1}{N}\sum_{m=0}^{N-1} \tilde{X}[m]\mathrm{e}^{\mathrm{j}\Omega_m k}$$

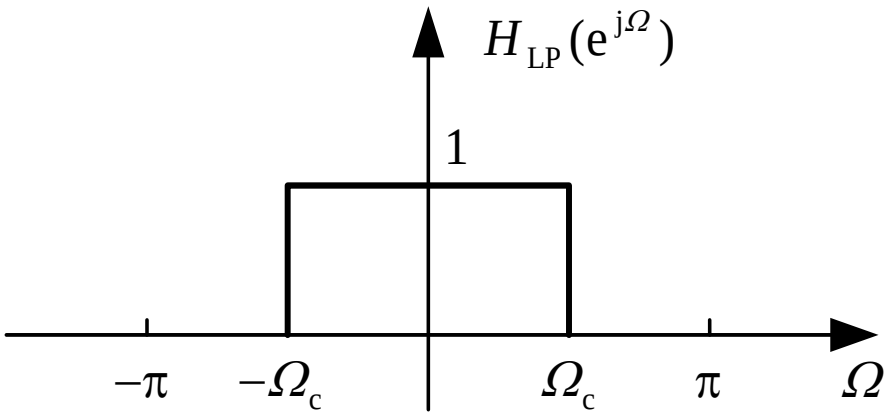
$$\left|H(\mathrm{e}^{\mathrm{j}\Omega})\right|\mathrm{e}^{-\mathrm{j}\Omega k_0}$$

$$\tilde{y}[k] = \frac{1}{N}\sum_{m=0}^{N-1} \tilde{X}[m]\left|H(\mathrm{e}^{\mathrm{j}\Omega_m})\right|\mathrm{e}^{\mathrm{j}\Omega_m (k-k_0)}$$

When a signal passes through a linear-phase LTI system, only the amplitude is changed and the phase is not distorted.

ideal digital filter

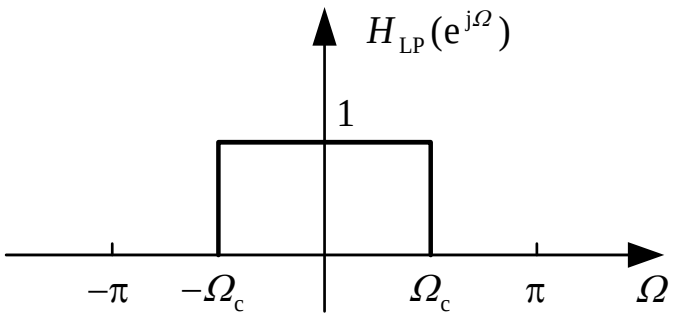
- ideal low pass filter

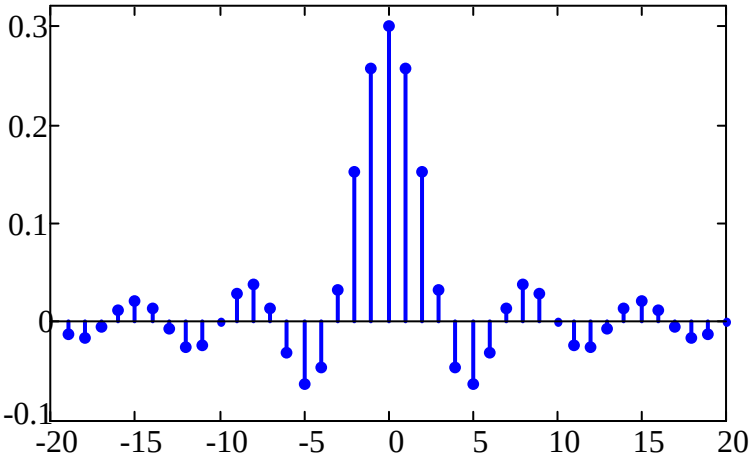


Frequency Response of an Ideal Low-Pass Filter

ideal digital filter

- Impulse Response of an Ideal Low-Pass Filter





Impulse Response of Low Pass Filter (Ω_c=0.3π)

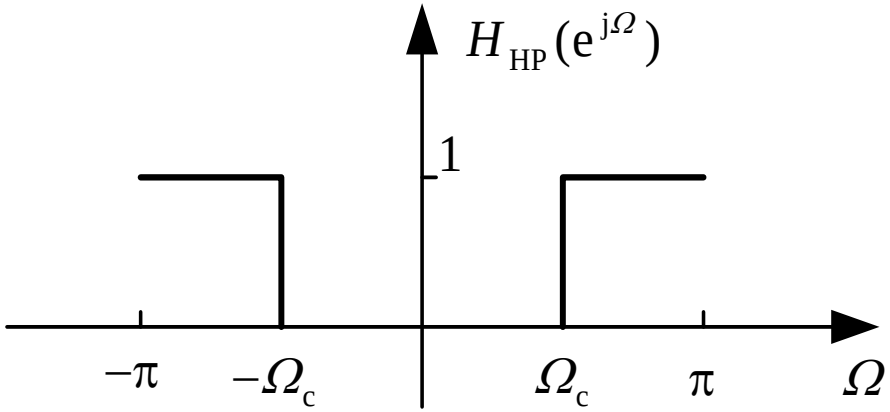
$$h_{LP}[k] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_{LP}(e^{j\Omega})e^{j\Omega k}d\Omega$$

$$= \frac{1}{2\pi}\int_{-\Omega_c}^{\Omega_c} e^{j\Omega k}d\Omega = \frac{\Omega_c}{\pi}\mathrm{Sa}(\Omega_c k)$$

An ideal digital low-pass filter is a non-causal system.

ideal digital filter

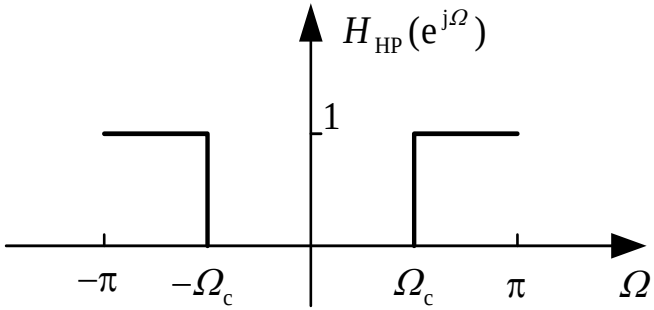
- ideal high pass filter

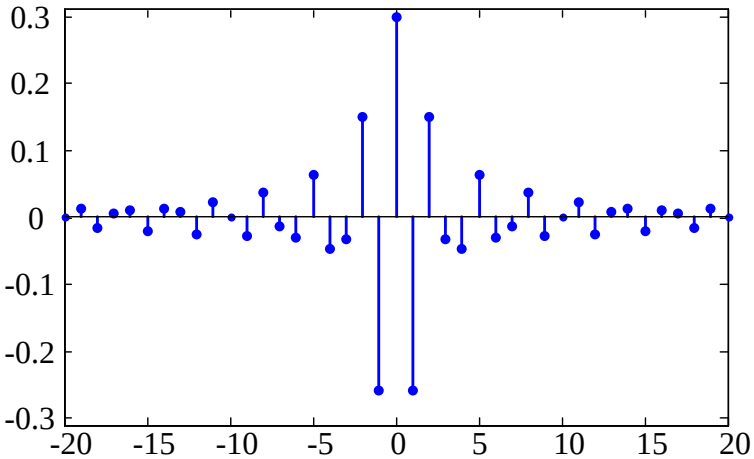


Frequency Response of an Ideal High-Pass Filter

ideal digital filter

- Impulse Response of an Ideal High-Pass Filter





Impulse Response of High Pass Filter (Ω_c=0.7π)

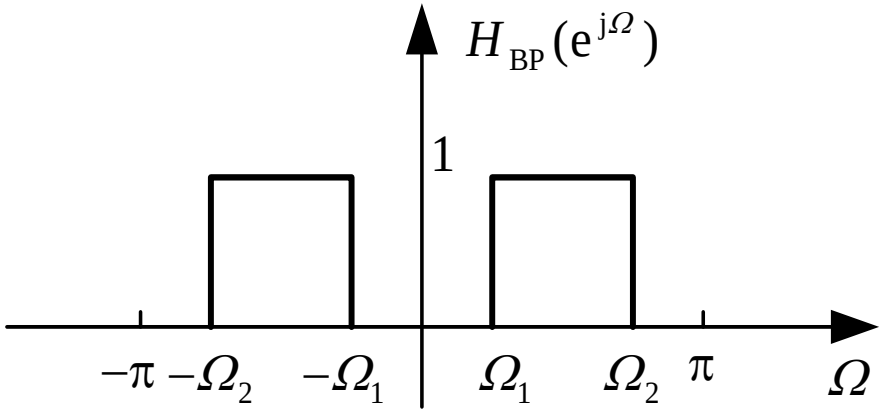
$$H_{HP}(e^{j\Omega}) = 1 - H_{LP}(e^{j\Omega})$$

$$h_{HP}[k] = \delta[k] - h_{LP}[k]$$

$$h_{HP}[k] = \delta[k] - \frac{\Omega_c}{\pi}\mathrm{Sa}(\Omega_c k)$$

ideal digital filter

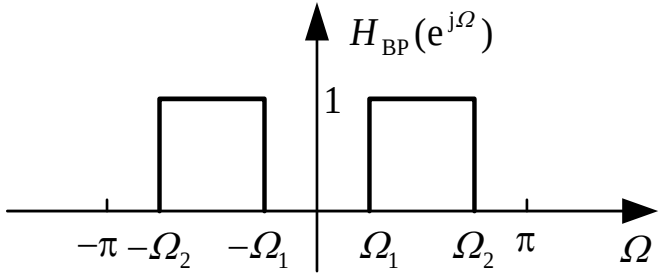
- ideal bandpass filter

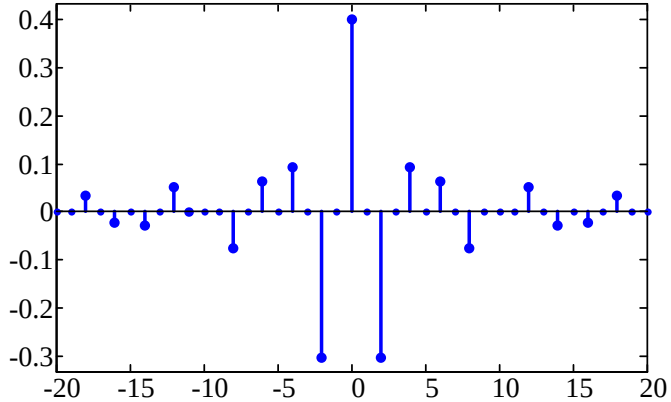


Frequency Response of an Ideal Bandpass Filter

ideal digital filter

- Impulse Response of an Ideal Bandpass Filter





Impulse Response of Bandpass Filter (Ω_2=0.7π， Ω_1=0.3π)

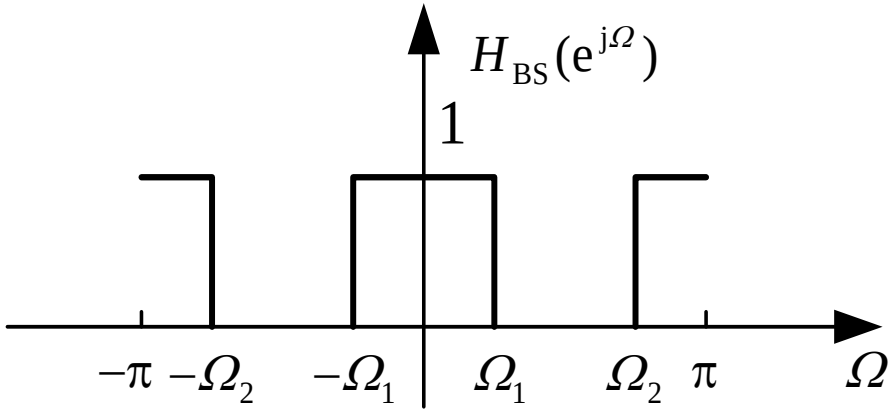
$$H_{BP}(e^{j\Omega}) = H_{LP2}(e^{j\Omega}) - H_{LP1}(e^{j\Omega})$$

$$h_{BP}[k] = h_{LP2}[k] - h_{LP1}[k]$$

$$h_{BP}[k] = \frac{\Omega_2}{\pi}\mathrm{Sa}(\Omega_2 k) - \frac{\Omega_1}{\pi}\mathrm{Sa}(\Omega_1 k)$$

ideal digital filter

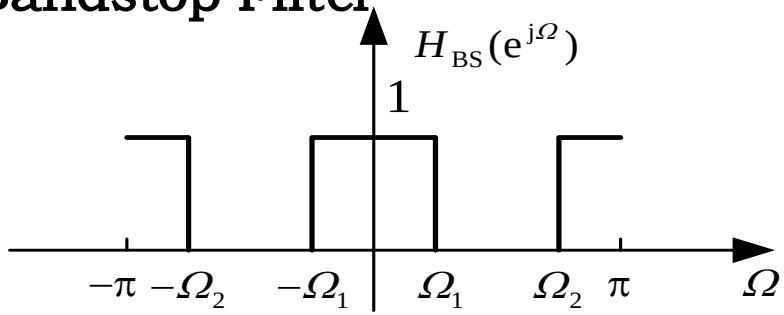
- ideal band stop filter

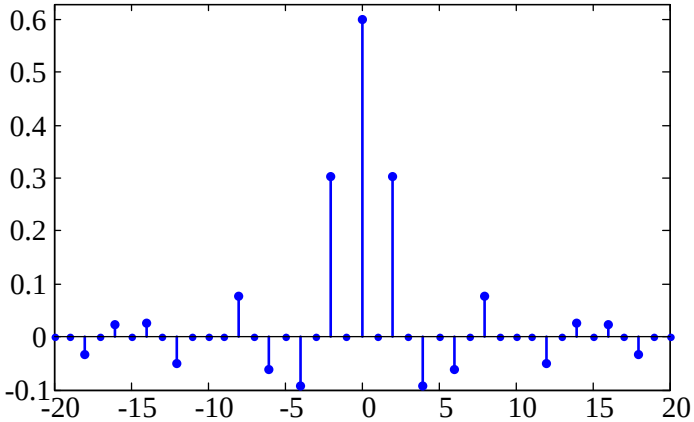


Frequency Response of an Ideal Bandstop Filter

ideal digital filter

- Impulse Response of an Ideal Bandstop Filter





Impulse Response of Bandstop Filter
(Ω_2=0.7π， Ω_1=0.3π)

$$H_{BS}(e^{j\Omega}) = H_{LP1}(e^{j\Omega}) + H_{HP2}(e^{j\Omega})$$

$$h_{BS}[k] = h_{LP1}[k] + h_{HP2}[k]$$

$$= \delta[k] + \frac{\Omega_1}{\pi}\mathrm{Sa}(\Omega_1 k) - \frac{\Omega_2}{\pi}\mathrm{Sa}(\Omega_2 k)$$

ideal digital filter

Example It is known that $H(\mathrm{e}^{\mathrm{j}\Omega})$ of an LTI system is

$$H(\mathrm{e}^{\mathrm{j}\Omega})=\begin{cases}\mathrm{e}^{-\mathrm{j}3\Omega} & |\Omega|<\dfrac{2\pi}{16}\dfrac{3}{2}\\ 0 & \text{其它}\end{cases}$$

Input as $\tilde{x}[k]=\tilde{\delta}_{16}[k]$ Find the output of the system.

Solution:

$$\tilde{\delta}_{16}[k]=\frac{1}{16}\sum_{m=0}^{15}\mathrm{e}^{\mathrm{j}\frac{2\pi}{16}km}$$

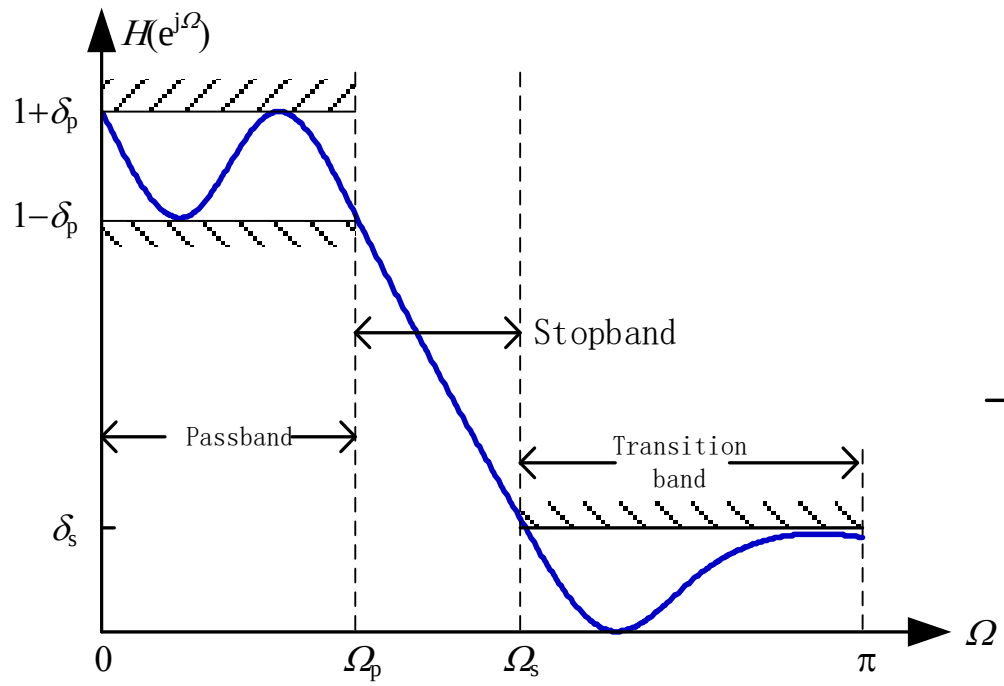
$$y[k]=\frac{1}{16}\{H(\mathrm{e}^{\mathrm{j}0})+H(\mathrm{e}^{\mathrm{j}\frac{2\pi}{16}})\mathrm{e}^{\mathrm{j}\frac{2\pi}{16}k}+H(\mathrm{e}^{\mathrm{j}\frac{2\pi}{16}15})\mathrm{e}^{\mathrm{j}\frac{2\pi}{16}15k}\}$$

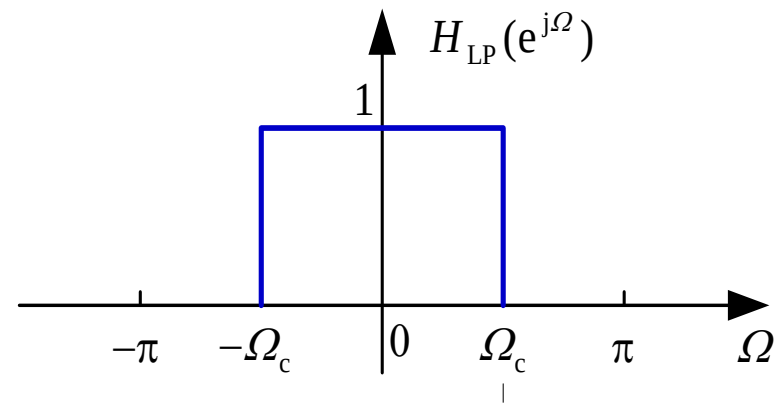
$$=\frac{1}{16}\{H(\mathrm{e}^{\mathrm{j}0})+H(\mathrm{e}^{\mathrm{j}\frac{2\pi}{16}})\mathrm{e}^{\mathrm{j}\frac{2\pi}{16}k}+H(\mathrm{e}^{-\mathrm{j}\frac{2\pi}{16}})\mathrm{e}^{-\mathrm{j}\frac{2\pi}{16}k}\}$$

$$=\frac{1}{16}\{1+2\cos[\frac{\pi}{8}(k-3)]\}$$

ideal digital filter

※ **Practical low-pass digital filter**





Ideal Low Pass Digital Filter

Frequency Domain Analysis of Discrete Time LTI Systems

- Advantages: When solving the zero-state response of the system, it can intuitively reflect the change of the signal spectrum after the signal passes through the system, explain the difference between the excitation and response time-domain waveforms, and the physical concept is clear.
- insufficient:

(1) Only the zero-state response of the system can be solved, and the zero-input response of the system

It still needs to be solved by the time domain method.

(2) If the Fourier transform of the system frequency response (or excitation signal) does not exist,

Then the frequency domain analysis method cannot be used.

- Solution: Use z-transform

Thanks

Some of the materials cited in this course are the teaching accumulation of the main lecturer for many years, which come from a variety of media and the exchanges of colleagues, peers and friends, and it is difficult to indicate the source one by one, so I hereby explain and thank you!